

# Notes: Daniel, Hirshleifer, and Subrahmanyam (JF, 1998)

## Investor Psychology and Security Market Under- and Overreactions

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# 1 Big picture

- **Question.** Can a single psychologically grounded theory explain why markets sometimes seem to underreact to public news and sometimes seem to overreact to prior information?
- **Core anomalies.** The paper targets several facts that were hard to reconcile with a purely rational efficient-market model: event-based drift, short-run momentum, long-run reversal, excess volatility, and post-earnings announcement drift followed by later reversal.
- **Two behavioral ingredients.** The theory uses (i) investor overconfidence about the precision of privately generated information and (ii) biased self-attribution, where confirming public feedback raises confidence more than disconfirming feedback reduces it.
- **Private versus public information distinction.** Investors are overconfident about private signals because they are personally involved in producing or interpreting those signals. They are not equally overconfident about public signals received by everyone.
- **Main price-path intuition.** Prices overreact to private signals, then partially correct when public signals arrive. With static overconfidence this creates long-run reversals. With biased self-attribution, confirming public signals initially intensify overreaction, producing short-run momentum and later reversal.
- **Event-study implication.** Public events predict future returns only when the event is *selective*, meaning its occurrence or magnitude is correlated with prior market mispricing. Nonselective public events can show event-date underreaction without systematic post-event drift.
- **Corporate-finance implication.** Managers and other informed parties choose financing, repurchase, split, or recommendation events in response to mispricing. This links investor psychology to corporate financing and payout behavior.
- **Economic takeaway.** The paper provides an integrated behavioral asset-pricing theory in which underreaction and overreaction are not contradictions. They are different phases of how overconfident investors process private and public information.

# 2 Incremental contribution of the paper

- **Unified behavioral explanation.** Earlier behavioral work often explained one anomaly at a time. Daniel, Hirshleifer, and Subrahmanyam provide a single framework for momentum, reversals, event drift, excess volatility, and corporate-event predictability.
- **Private-public signal asymmetry.** A key contribution is distinguishing investor overconfidence about privately generated signals from beliefs about public signals. This distinction lets the model generate both overreaction and underreaction.
- **Dynamic confidence updating.** The paper adds biased self-attribution to make confidence time varying. This converts pure overreaction-correction into a richer pattern with short-run continuation and long-run reversal.
- **Selective versus nonselective events.** The paper clarifies that post-event drift is not automatic evidence of underreaction. Drift should be concentrated in events selected by informed parties because they are correlated with mispricing.
- **Event-study predictions.** The theory generates testable predictions about voluntary versus involuntary events, event size, pre-event runups, market/book ratios, and long-run post-event performance.
- **Behavioral foundation from psychology.** Rather than assuming arbitrary irrationality, the model is built on two well-documented psychological regularities: overconfidence and self-attribution bias.
- **Link to corporate policy.** The paper connects market mispricing to managerial actions such as repurchases, equity issuance, stock splits, analyst recommendations, and earnings management.

- **Cultural implication.** Because self-attribution varies across cultures, the model predicts cross-country variation in momentum and post-earnings drift, an implication that was relatively unusual for asset-pricing theory at the time.
- **Conceptual innovation.** The paper shows that positive autocorrelation need not mean underreaction and negative autocorrelation need not be the only form of overreaction. Positive short-run autocorrelation can itself arise from continuing overreaction.

## 3 Model objects and measurement

### 3.1 Information signals

- The true value of the security is denoted by  $\theta$ .
- At date 1, informed investors receive a private signal, often written as  $s_1 = \theta + \varepsilon$ .
- At later dates, public signals arrive, such as  $s_2$  or a sequence of public signals  $\phi_t = \theta + \eta_t$ .
- Rational investors weight signals according to their true precisions. Overconfident investors overestimate the precision of private signals.

**Interpretation:** The model is about relative signal weighting. Overconfident investors do not ignore public news; they overweight private news relative to public news.

### 3.2 Prices and mispricing

- $P_t$  is the equilibrium security price at date  $t$ .
- A fully rational benchmark price weights signals using true variances.
- Mispricing is the gap between the overconfident-investor price and the rational benchmark price.
- Price changes, such as  $P_1 - P_0$  or  $P_2 - P_1$ , are the objects used to study momentum, reversal, and event drift.

**Interpretation:** Mispricing is dynamic: a private signal can push price away from value, while public signals gradually correct or amplify that error.

### 3.3 Overconfidence

- Overconfidence is modeled as an overestimation of private signal precision.
- In the paper's interpretation, investors believe they are better at security valuation than they actually are.
- This is most plausible for difficult, judgment-intensive tasks with delayed feedback, such as valuing growth firms or intangible assets.

**Interpretation:** Overconfidence gives a disciplined behavioral reason for why market prices can overreact specifically to private information.

### 3.4 Biased self-attribution

- If later public information confirms an investor's prior private signal, confidence rises.
- If later public information disconfirms the private signal, confidence falls only weakly, if at all.
- On average, this asymmetry causes confidence to drift upward after some confirmations and creates continuing overreaction.

**Interpretation:** Self-attribution turns a static overreaction story into a dynamic story that can produce momentum before reversal.

### 3.5 Events

- A **nonselective event** is unrelated to prior market mispricing.
- A **selective event** is chosen or sized by an informed party in response to prior mispricing.
- Corporate events such as repurchases, equity issuance, splits, or recommendations can be selective if informed managers or analysts act strategically.

**Interpretation:** The selective/nonselective distinction is the paper's most important event-study contribution. Public news and corporate actions are not all economically equivalent.

## 4 Models and theoretical specifications

### 4.1 Basic signal-weighting structure

A stylized version of the model can be summarized as follows:

$$P_1 = E^{OC}[\theta \mid s_1], \quad P_2 = E^{OC}[\theta \mid s_1, s_2], \quad P_3 = E[\theta \mid s_1, s_2, s_3].$$

Here  $E^{OC}$  denotes valuation using overconfident precision for the private signal. The rational benchmark uses true precision.

**Interpretation:** The date 1 price overweights the private signal. Later public signals push the price back toward the rational value, unless self-attribution causes confidence to rise further.

### 4.2 Overreaction and correction

The basic overconfidence model predicts:

$$\text{cov}(P_2 - P_1, P_1 - P_0) < 0,$$

$$\text{cov}(P_3 - P_1, P_1 - P_0) < 0,$$

$$\text{cov}(P_3 - P_2, P_2 - P_1) > 0.$$

**Interpretation:** The first two inequalities are reversal effects after private-signal overreaction. The third inequality says that correction following public information can generate positive correlation between public-news price changes and subsequent price changes.

### 4.3 Excess volatility

The paper defines proportional excess volatility around date  $t$  as:

$$VE_t = \frac{\text{var}(P_t - P_{t-1}) - \text{var}_R(P_t - P_{t-1})}{\text{var}_R(P_t - P_{t-1})}.$$

**Interpretation:** Overconfidence increases volatility around private signals and increases unconditional volatility, but the effect around public signals is ambiguous because public news partly corrects prior overreaction.

### 4.4 Selective-event pricing

Let  $e^*$  denote the date 2 pricing error inferred by an informed manager. If  $e^* < 0$ , the market undervalues the firm; if  $e^* > 0$ , the market overvalues the firm. The model implies:

$$E[P_3 - P_2 \mid e^* > 0] < 0 < E[P_3 - P_2 \mid e^* < 0].$$

**Interpretation:** Events chosen in response to undervaluation, such as repurchases, should be followed by positive abnormal performance. Events chosen in response to overvaluation, such as equity issuance, should be followed by negative abnormal performance.

### 4.5 Dynamic self-attribution model

With a sequence of public signals, self-attribution makes perceived private-signal precision evolve over time. Confirming public signals raise perceived precision more than disconfirming public signals lower it.

$$\text{Confidence}_{t+1} = F(\text{Confidence}_t, \text{confirmation of } s_1 \text{ by public signals}).$$

**Interpretation:** This creates an average price path that initially moves away from fundamentals after a private shock and later reverts as enough public evidence accumulates.

## 5 Identification and theoretical design logic

### 5.1 What plays the role of identification

This is a theoretical paper rather than a direct empirical test. The identification logic is internal: the model maps distinct psychological mechanisms to distinct return patterns.

- Static overconfidence generates overreaction to private signals and long-run reversal.
- Public-signal underreaction generates positive event-date and post-event return correlation.
- Biased self-attribution generates short-run momentum and later reversal.
- Selective events generate event-based drift; nonselective events do not.

### 5.2 Why the model is disciplined

- The psychology is not arbitrary; it is based on documented overconfidence and attribution biases.
- The theory makes predictions outside the original anomalies, such as stronger momentum in hard-to-value growth firms and weaker momentum in cultures with less self-enhancing attribution.
- The theory differentiates voluntary corporate actions from involuntary or nonselective public events.

### 5.3 What the theory does not do

- It does not estimate behavioral parameters directly.
- It does not provide a full general-equilibrium model of arbitrage capital and survival.
- It does not claim that all anomalies must come from one bias.
- It relies on limits to arbitrage or persistence of quasi-rational investors for mispricing to affect prices.

## 6 Section-by-section study guide

- **Introduction.** Motivates the need for an integrated behavioral theory of underreaction and overreaction.
- **Section I.** Reviews psychological evidence on overconfidence and biased self-attribution.
- **Section II.** Builds the basic overconfidence model and derives price-path, autocorrelation, volatility, and event-study implications.
- **Section III.** Introduces dynamic biased self-attribution and shows how it generates momentum, long-run reversal, and earnings-drift implications.
- **Section IV.** Discusses survival of overconfident traders, links to noise-trading theories, and summarizes empirical implications.
- **Appendices.** Catalogue anomalies, provide covariance calculations, develop a discrete self-attribution model, and provide dynamic covariance calculations.

## 7 Figures and theoretical result blocks

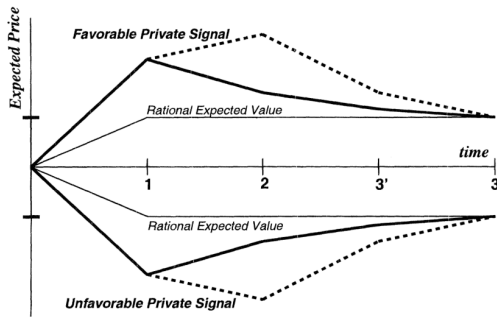
### 7.1 Figures 1 and 2: Average price paths under overconfidence and self-attribution

Read:

- Figure 1 shows average prices after favorable and unfavorable private signals in models with and without self-attribution bias.
- The solid path illustrates static overconfidence: prices jump too far after private information and then correct toward rational value.
- The dashed path illustrates self-attribution: confirming public information can push prices further away from fundamentals before eventual correction.
- Figure 2 simulates a private information shock and compares the average path with and without attribution bias.

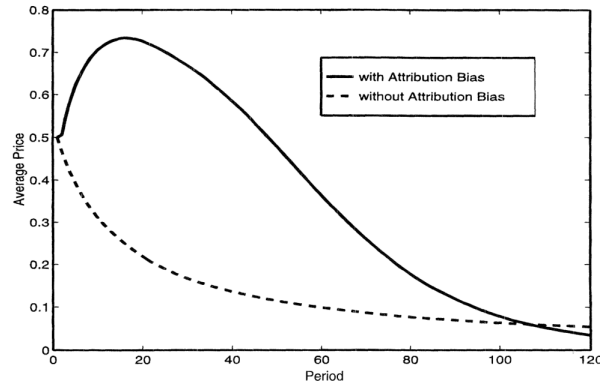
**Detailed interpretation:** Figure 1 is the conceptual map of the paper. The market overreacts to a private signal because the investor overweights it. Public information then has two roles. First, it is rational evidence that eventually corrects the price. Second, under self-attribution, early confirming public signals raise confidence and produce continuing overreaction. Figure 2 turns this logic into a longer simulated path: the attribution-bias price path rises for many periods before the eventual decline, while the no-attribution path declines steadily.

**Economic message:** The same private signal can generate both momentum and reversal. Momentum is not simple underreaction; it can be continuing overreaction.



**Figure 1. Average price as a function of time with overconfident investors.** This figure shows price as a function of time for the dynamic model of Section III with (dashed line) and without (solid line) self-attribution bias.

(a) Figure 1: Average price as a function of time with overconfident investors.



**Figure 2. Average price path following private information shock.** This figure shows the average price path calculated using the simulation in Section III.B.3, following a private information shock  $s_1 = 1$ . The price path is shown for the dynamic model with (solid line) and without (dashed line) self-attribution bias.

(b) Figure 2: Average price path following private information shock.

### 7.2 Figures 3 and 4: Autocorrelations and information-change correlations

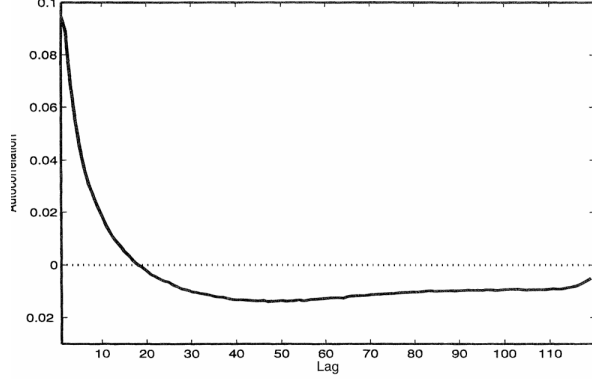
Read:

- Figure 3 reports simulated average price-change autocorrelations at lags from 1 to 119 periods.
- Short-lag autocorrelations are positive, while longer-lag autocorrelations become negative.
- Figure 4 reports correlations between information changes and future price changes.
- It shows positive short-lag correlations but negative or attenuated longer-lag correlations.



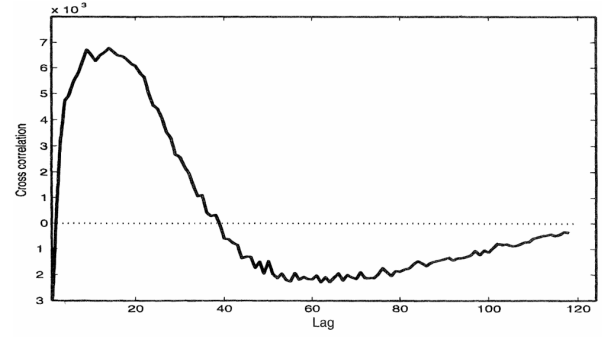
**Detailed interpretation:** Figure 3 visually delivers the model’s main dynamic prediction: short-run momentum coexists with long-run reversal. Figure 4 connects that dynamic to accounting or public-information surprises. Short-run public surprises can predict same-direction future returns because they stimulate overconfident belief reinforcement. Over longer horizons, the price correction can dominate.

**Economic message:** The model explains why post-earnings announcement drift and long-run reversal can coexist. Public news can initially reinforce overreaction but eventually help correct it.



**Figure 3. Average price change autocorrelations.** This figure presents the unconditional average autocorrelations (at lags between 1 period and 119 periods), calculated using the simulation described in Section III.B.3.

(a) Figure 3: Average price change autocorrelations.



**Figure 4. Correlation between information changes and future price changes.** This figure shows the set of average sample correlations between the  $\Delta e_t$  and price changes  $\tau$  periods in the future  $\Delta P_{t+\tau} = P_{t+\tau} - P_{t+\tau-1}$ . These are calculated using the simulated dynamic model of Section III.B.3.

(b) Figure 4: Correlation between information changes and future price changes.

### 7.3 Theoretical result block: Propositions 1–3

#### Propositions 1–3: Static overconfidence

- **Proposition 1:** Price moves due to private information are partially reversed in the long run; price moves due to public information are positively correlated with later price changes.
- **Proposition 2:** In the constant-confidence model, unconditional price changes are negatively autocorrelated at short and long lags.
- **Proposition 3:** Overconfidence increases volatility around private signals, can increase or decrease volatility around public signals, and increases unconditional volatility.

**Interpretation:** Static overconfidence is enough to generate long-run reversal and excess volatility. It is not enough to generate short-run momentum in the main specification. This is why the paper adds biased self-attribution.

## 7.4 Theoretical result block: Propositions 4–6

### Propositions 4–6: Event-study implications

- **Proposition 4:** For nonselective events, true expected post-announcement abnormal return is zero, even though announcement and post-announcement price changes can be positively correlated conditional on pre-event returns.
- **Proposition 5:** Selective events undertaken when stocks are undervalued predict positive announcement and post-announcement returns; selective events undertaken when stocks are overvalued predict negative announcement and post-announcement returns.
- **Proposition 6:** The size and probability of selective events vary with measures of mispricing such as market/book, price/earnings, or prior runup.

**Interpretation:** Event-based drift is not just underreaction to public news. Drift arises when an event is selected because of mispricing. This distinguishes repurchases, equity issues, analyst recommendations, and other strategic events from exogenous public news.

## 7.5 Theoretical result block: Proposition 7 and dynamic results

### Proposition 7 and dynamic self-attribution

- **Proposition 7:** If confidence changes due to biased self-attribution, short-lag autocorrelations can be positive while long-lag autocorrelations are negative.
- **Result 1:** After a favorable private signal, the price can first rise further and then decline.
- **Result 2:** Short-lag price-change autocorrelations are positive, but long-lag autocorrelations are negative.
- **Result 4:** Short-lag correlations between price changes and accounting surprises are positive, while longer-lag correlations can turn negative.

**Interpretation:** Biased self-attribution is the model's momentum engine. It produces a short-run continuation phase before the long-run correction phase.

## 8 Detailed result map

- **Overconfidence about private signals:** private news is overweighted, generating initial overreaction.
- **Public-signal correction:** public news gradually corrects private-signal overreaction.
- **Static model:** generates long-run reversals and excess volatility, but not robust short-run momentum.
- **Self-attribution:** confirming public information raises confidence and creates continuing overreaction.
- **Momentum and reversal:** prices can continue in the short run and reverse in the long run.
- **Event studies:** selective events predict drift; nonselective events do not systematically predict future returns.
- **Corporate policy:** firms may issue equity when overpriced and repurchase when underpriced.
- **Cross-sectional implications:** momentum should be stronger for hard-to-value stocks and potentially weaker in cultures with less self-enhancing attribution.

## 9 Short summary

- The paper offers a behavioral theory of underreaction and overreaction.
- Investors are overconfident about privately generated information but not public signals.
- Overconfidence creates overreaction to private signals and later correction through public information.
- Biased self-attribution makes confidence rise after confirming public feedback and fall little after disconfirming feedback.
- Self-attribution generates short-run momentum and long-run reversal.
- Event drift is predicted mainly for selective events chosen in response to mispricing.
- The theory connects return anomalies to corporate financing and payout actions.
- The paper is theoretical and has four main figures but no empirical tables.

## 10 Conclusion

### 10.1 Core conclusion

Daniel, Hirshleifer, and Subrahmanyam develop an integrated behavioral asset-pricing theory in which overconfidence and biased self-attribution jointly explain underreaction, overreaction, momentum, long-run reversal, excess volatility, post-earnings drift, and event-based return predictability.

### 10.2 Economic intuition

Investors overestimate the precision of private information that they generate themselves. This causes prices to move too much when private information is incorporated. Public information should correct the error, but if it initially confirms the investor's private signal, self-attribution raises confidence and causes additional price movement in the same direction. Eventually, enough public information arrives to correct the mispricing.

### 10.3 Incremental contribution revisited

The paper's key incremental contribution is to show that a psychologically grounded model can explain both apparent underreaction and apparent overreaction. The model is parsimonious: overconfidence creates private-signal overreaction, while self-attribution creates continuing overreaction that looks like momentum. The selective-event framework then connects this price psychology to corporate event anomalies.

### 10.4 Implications for empirical work

The paper suggests that researchers should distinguish private-information events from public-information events, selective events from nonselective events, short-run continuation from long-run reversal, and hard-to-value stocks from easy-to-value stocks. It also predicts that corporate actions such as equity issues, repurchases, and recommendations should be more informative when they are chosen in response to mispricing.

### 10.5 Limitations

The paper does not directly estimate behavioral parameters or provide new empirical tests. It also relies on the persistence of overconfident traders despite arbitrage. Its strength is conceptual integration and the

generation of cross-context empirical implications.

## **10.6 Takeaway**

The main lesson is that underreaction and overreaction are not mutually exclusive market behaviors. They can be different phases of one psychological price process: overreaction to private information, temporary reinforcement by biased self-attribution, and eventual correction by public information.